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## The Completion

**\*\*A relativistic field theory carrying  $a_0 = \kappa c \sqrt{(G\rho - \Lambda)}$ \*\***

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2 same day**. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line. **v2 changes:** row 13's shift-charge cluster route is **withdrawn** (killed by the 1-Mpc confrontation: smooth accretion restores  $\$ \rightarrow 1 \text{ for any cold IC) and replaced by the } *a_{-}\{0\} - \text{bump environment response}$ , which passed a five—environment matrix AND its perturbation health check  $\leq 8.4 \times 10^{-7}$ ; amplitude  $\leq 2.7 \times$  fiducial). The coefficient row now reads as **measured:**  $\kappa = 0.551 \pm 0.043$ ,  $\pm 0.063$  if the  $H_0$  tension is carried as a systematic ( $\kappa \propto a_0/H_0$ ).

### 1. The action

$$S = \int d^4x \sqrt{-g} \left[ \underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein-Hilbert}} + \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} + \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}} \right] + S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon  $\varphi$  and the aether  $A^\mu$ ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi.$$

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework's content is entirely in the choice of the free function , which **factorises**:

$$\mathcal{F}(Y, Q) = \underbrace{\frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} + \underbrace{K(Q)}_{\text{dark energy + dark matter}} + \underbrace{A B\left(\frac{Y}{a_0^2}\right) (Q - Q_0)^2}_{a_0\text{-bump response (v2)}}$$

$$B(y) = \frac{y}{(1 + y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass**  $\mu_{\text{eff}}^2 = A B(g^2/a_0^2)$  **peaked at the framework's own**  $a_0$  — the location costs nothing ( $Y/a_0^2$  is already the Y-sector's normalisation); one new calibrated amplitude  $A \approx 1.7 \text{ Mpc}^{-2}$ . *The dark sector's response is resonant at the MOND transition, and clusters are the cosmic objects that live at  $a_0$  ( $R_{500}$  at  $0.33\text{--}0.58 a_0$ ).*

## 1.1 The Y-sector — MOND

$\_Y$  is fixed by the framework's Route A kernel  $\nu(y) = 1/(1 - e^{\widehat{(-\sqrt{y})}})$ ,  $y = g\_bar/a_0$ . In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \frac{1}{2}, \quad Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$  is **FITTED, not derived**. See §4.

## 1.2 The Q-sector — an offset DBI khronon

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[ 1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D$$

Three jobs from one function:

| piece                    | limit                      | what it is                                                                                             |
|--------------------------|----------------------------|--------------------------------------------------------------------------------------------------------|
| the <b>offset</b> $-M^4$ | $u = 0, K = 0$             | $\rho = -K = M^4, p = K = -M^4$<br>$\Rightarrow \mathbf{w} = \mathbf{-1}$ <b>exactly</b> : dark energy |
| the <b>deviations</b>    | $u \ll \Lambda\_D$         | $K \rightarrow \mu^2 u^2/2 \Rightarrow w \rightarrow 0$ : <b>dust</b> ,<br>i.e. the dark matter        |
| the <b>saturation</b>    | $u \rightarrow \Lambda\_D$ | $K$ bounded, $K \rightarrow \infty \Rightarrow w \rightarrow 0$ again: kills the early stiff phase     |

Parameters:  $\mathbf{M}$  (vacuum scale,  $M^4 = \rho\_ \Lambda$ ),  $\mu$  (Helmholtz mass),  $\mathbf{\Lambda\_D}$  (DBI field-space scale),  $\mathbf{I}_0$  (the conserved shift charge = the dust amount).

## 2. What is verified

| # | property                                               | value / statement                                                                                                                                                           | script                                        |
|---|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1 | <b>Lensing</b> — the test that killed modified inertia | $\Phi = \Psi \Rightarrow \gamma\_ \text{PPN} = 1, M\_ \text{dyn}/M\_ \text{lens} = 1$ exactly. <b>21.2<math>\sigma</math></b> $\rightarrow$ <b>0.601<math>\sigma</math></b> | <code>mi_relativistic_completion_aest.</code> |

| #  | property                                                | value / statement                                                                                                                                                                                                                                                                                                     | script                           |
|----|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 2  | <b>Kernel embeds</b> in<br>_Y                           | deep-MOND $\mu \rightarrow x$<br>exact; Newtonian<br>$\$ \rightarrow 1; x(u) \text{abijection}(h =$<br>$e^{(-u)}(1+u) >$<br>$0); \text{free function} *$<br>$* \text{convex} **   \text{same}    3  *$<br>$* \text{Newtonian residual} *$<br>$*   e^{(-\sqrt{\$y})} =$<br>$3.6 \times 10^{-3457}$ at<br>Earth's orbit | same                             |
| 4  | <b>Dark energy is not</b><br><b>an input</b>            | w = -1 <b>exactly</b> at the<br>condensate minimum<br>— proved, not<br>assumed                                                                                                                                                                                                                                        | mi_condensate_vacuum_energy_a0_2 |
| 5  | <b>Ghost-free</b> over the<br>whole field range         | $K = \mu^2(1-s^2)^{-3/2} >$<br>0 for all $ s  < 1$                                                                                                                                                                                                                                                                    | mi_dbi_khronon_2026.py           |
| 6  | <b>Subluminal</b><br>everywhere                         | $c_{s^2} = \Lambda\_D$<br>$s(1-s^2)/(1+\Lambda\_D s) \leq$<br>$0.385 \Lambda\_D$                                                                                                                                                                                                                                      | same                             |
| 7  | <b>The published</b><br><b>no-go is dissolved</b>       | Blanchet & Skordis<br>2024 §4.3.1's $455 \times$<br>conflict <b>reverses</b><br><b>sign</b> : bounded<br>pressure $\Rightarrow w \$ \rightarrow \$0$<br>early $\Rightarrow$ cosmology<br>becomes a <i>lower</i><br>bound on $\mu^{-1}$ , same<br>direction as MOND                                                    | same                             |
| 8  | <b>Theorem</b> : why no<br>polynomial works             | for $K \sim u^n$ , early-time<br>$w \rightarrow 1/(n-1)$ .<br>Quadratic 1, quartic<br>$\frac{1}{3}$ , sextic — all fail.<br><b>Only boundedness</b><br><b>gives w\$→\$0</b>                                                                                                                                           | same                             |
| 9  | <b>CMB acoustic</b><br><b>peaks</b> — real CLASS<br>run | at $\Lambda\_D \leq 10^{-2}$ ,<br>$c_{s^2}(\text{recomb}) =$<br>$2.9 \times 10^{-8}$ ; TT<br>unchanged to <b>0.069%</b> ,<br>P(k=0.2) to 1.71%.<br>Indistinguishable from<br>CDM                                                                                                                                      | mi_dbi_cmb_class_run_2026.py     |
| 10 | <b>BTFR</b>                                             | exact, from convexity                                                                                                                                                                                                                                                                                                 | mi_route_a_field_theory_2026.py  |
| 11 | <b><math>g^{-2}</math> Lorentz</b><br><b>violation</b>  | <b>restored</b> by the<br>aether (pure<br>Bekenstein–Milgrom<br>had lost it)                                                                                                                                                                                                                                          | mi_relativistic_completion_aest. |

| #  | property                                          | value / statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | script                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | <b>a<sub>0</sub> tied to <math>\Lambda</math></b> | $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$<br>with $m_{\text{cond}} = M^2/(\sqrt{2}M_{Pl}), M^4$<br>$= \rho_{\Lambda}$ — agrees to<br><b>0.076%</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | mi_condensate_vacuum_energy_a0_2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 13 | <b>Clusters (v2)</b>                              | <del>shift-charge-IC route</del><br><b>withdrawn:</b> killed by<br>the 1-Mpc<br>confrontation —<br>cluster R500 and<br>galaxy-outskirt lensing<br>share a scale with<br>disjoint requirements,<br>correlated Gaussian<br>ICs cannot select<br>environments, and<br><b>smooth accretion</b><br><b>restores</b><br>$\$ \rightarrow 1$ for any cold IC *<br>*.Replaced by the *<br>*a{0}-bump response*<br>*(row15) \mi{c_route_1mpc}untouched by n_2026.py` (9/9)  14 *<br>*Directional EFE signal*<br>* pure MI predicted *<br>exactly zero*; AQUAL—<br>class predicts 1–4 15  *<br>*a{0}\$-bump<br><b>environment</b><br><b>response</b> | passes all five<br>environments: cluster<br>(calibration, $\mu^2_{\text{eff}} =$<br>0.23 Mpc <sup>-2</sup> ), galaxy<br>interior 0.034% of<br>M <sub>b</sub> (RAR cost<br>$4 \times 10^{-4}$ dex), galaxy<br>1 Mpc 1.2% vs the<br>5.9% strict bound,<br>linear cosmos 0.6% of<br>mean matter, solar<br>$10^{-19}$ . Vanishes on<br>FRW (Y=0) and is<br>second-order in<br>perturbations $\Rightarrow$<br>linear CMB/P(k)<br>construction. Evades<br>both prior kills (a<br>response has no charge<br>to advect; couples to<br>environment directly) |

| #  | property                       | value / statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | script                               |
|----|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 16 | <b>...and its health check</b> | <b>no ghosts ever</b><br>(kinetic contribution<br>$2AB \geq 0$ , $= 0$ on<br>FRW), no<br>Ostrogradsky sector;<br>FRW gradient health<br>**excludes the old<br>$\Lambda\_D = 10^{-2}$ and<br>forces $\Lambda\_D \leq$<br>$8.4 \times 10^{-7}$ ** (window<br>stays 3.7 orders); halo<br>gradient health <b>caps</b><br><b><math>A \leq 2.7 \times</math> fiducial,</b><br><b>excluding the</b><br><b>demanding end of</b><br><b>Mistele's cluster</b><br><b>band</b> — a two-sided<br>falsifiable pinch; free<br>signature: cluster<br>anisotropic stress at<br>$O(0.6)$ | mi_a0_bump_health_2026.py<br>(11/11) |

### 3. Parameter count, honestly

|             | $\Lambda$ CDM              | this completion                                                                                                                                                                                                                                                                                                                                     |
|-------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dark sector | $\Omega_c, \Omega_\Lambda$ | $M (= \rho_\Lambda^{-1/4})$ , $I_0$ , $\mu$ ,<br>$\Lambda\_D$ , <b>A (v2)</b>                                                                                                                                                                                                                                                                       |
| MOND scale  | —                          | $a_0$ — <b>not independent</b> , =<br>$m_{\text{cond}}/(4\sqrt{\pi})$ via item 12                                                                                                                                                                                                                                                                   |
| bounds      | —                          | $\Lambda\_D$ : $1.9 \times 10^{-10} \ll \Lambda\_D \leq$<br>$8.4 \times 10^{-7}$ ( <b>health-bounded,</b><br><b>v2</b> ; 3.7 orders); $A \approx 1.7$<br>$\text{Mpc}^{-2}$ calibrated on clusters,<br><b>health cap <math>A \leq 4.5</math></b> (row 16);<br>$\mu^{-1} = 4392 \text{ Mpc}$ (item 12); $I_0$<br>$\approx \Omega_{\text{dm}}$ (an IC) |

So: **five dark-sector numbers against  $\Lambda$ CDM's two** (v2 adds  $A$ ), with  $a_0$  derived from one of them rather than added. Not fewer parameters. What it buys is that  $\Lambda$ , dark matter and MOND come from **one function** instead of three unrelated sectors.

#### 4. What is NOT claimed — read this before quoting anything above

1.  $\kappa = \frac{1}{2}$  **IS NOT DERIVED**. Item 12 looks like a derivation and is not.  $a = m_{\text{cond}}/(4\sqrt{G\Lambda})$  is *algebraically identical* to  $a = \frac{1}{2}\sqrt{G\Lambda}$  and to  $Z = 2\sqrt{8/3}$  — three costumes, one statement — and the residue is **convention-dependent** ( $4\sqrt{\pi}$  with the reduced Planck mass,  $\sqrt{2}$  with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus’s own theorem already established. What genuinely moved: “why  $\kappa = \frac{1}{2}$ ?” becomes “why is  $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ ?”, which is sharper because  $m_{\text{cond}}$  is a *derived* quantity of a real field theory rather than a definition.
2. **Dark matter EXISTS**, at the full  $\Omega_{dm}$ . *Removing the pressureless component moves  $H_3/H_1$  by 54% and no refit absorbs it ( $\Delta\chi^2 > 400$ ). What does **not** exist is a dark-matter particle: it is the  $Q$ -sector of the same scalar that supplies  $\Lambda$  and MOND, and by item 13 it is absent where rotation curves are measured. The defensible slogan is “no dark matter particle, and none in galaxies” — never “no dark matter.”* 2b. **The coefficient is MEASURED, not derived:**  $\kappa = 0.551 \pm 0.043$  (distance-free SPARC estimator), and  $\pm 0.063$  if the  $H_0$  tension is carried as a systematic, since  $\kappa \propto a_0/H_0$  at full strength.  $\frac{1}{2}$ ,  $1/\sqrt{3}$ ,  $\sqrt{3/8}$  and  $0.40$  all sit inside  $2\sigma$ . (*v2 correction: v1’s  $\pm 0.043$  omitted the  $\rho\Lambda$  term.*) 2c. **The  $a_0$ -bump response is a CANDIDATE, not a result:** it has passed the environment matrix and its isolated-term health check; the full AeST perturbation matrix (aether mixings,  $K_B$  terms) and a real cluster model remain owed, and the health-vs-Mistele amplitude pinch could still kill it.
3. **The post-recombination growth history is NOT verified**. Item 9 covers the acoustic peaks only. Holding  $c_s^2$  at its peak value for all time *destroys*  $P(k=0.2)$ , so the pass depends essentially on the bump being transient (it peaks at  $z \approx 189$ ). **A patched CLASS fluid carrying  $c_s^2(a)$  is required, not optional.**
4. **Item 13’s initial condition is not confronted with Lyman- $\alpha$ .** It needs a khronon transfer function falling to  $T \approx 0.33$  at  $k \approx 4.5 \text{ Mpc}^{-1}$ , which in a  $\Lambda$ CDM-style analysis would be excluded. Whether the MOND  $Y$ -sector’s compensation rescues it is **uncomputed, and is the largest single owed item in the programme.**
5. **Cassini  $Q_2$  is inherited** ( $3\text{--}15\sigma$ ) and is not relieved by anything here.
6. **PPN preferred-frame parameters  $\alpha_1, \alpha_2$**  for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.**  $\gamma_v = \sqrt{\nu(y_{\text{extN}})}$ ; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

#### 5. The falsifiable predictions

| prediction                | value                                    | how to test                                       |
|---------------------------|------------------------------------------|---------------------------------------------------|
| wide-binary boost         | $\gamma_v = \sqrt{\nu(y_{\text{extN}})}$ | <b>Gaia DR4</b> , pre-registered and hash-stamped |
| directional EFE asymmetry | 1–4%, signed                             | already firing at $p = 0.029$ with the right sign |

| prediction                                        | value                                                             | how to test                                                                                              |
|---------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| cluster-to-cluster <b>scatter</b> in the residual | $\sigma(\xi)/\xi \approx 0.02\text{--}0.60$                       | existing cluster samples — <b>separates item 13 from the <math>R^2</math> lever, which predicts zero</b> |
| cluster mass <b>profile</b> misfit                | $\rho_{\text{c}}$ flat, $M_{\text{c}} \sim r^3$ , can go negative | fit clusters with the khronon profile instead of NFW                                                     |
| $g^{-2}$ Lorentz violation                        | computable $s_{\mu\nu}$                                           | SME sector bounds                                                                                        |
| accelerated structure formation                   | earlier massive objects                                           | JWST high-z massive galaxies (McGaugh et al. 2024)                                                       |

## 6. Credit

$\nu = \sqrt{1+1/y}$  is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złóśnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036.  $K(Q) = \mu^2(Q-1)^2$  and the ghost- condensate identification: Verwayen, Skordis & Złóśnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing challenge item 13 resolves. GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly- $\alpha$  floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

\*\*The distinctive content of this work is the coefficient —  $a_0 = \kappa c \sqrt{G\rho_{\text{c}}\Lambda}$  with  $\kappa = \frac{1}{2}$  — its embedding in  $\_Y$ , the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.\*\*